

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT3 — TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Time-Series & Geospatial Analysis Report
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO4 Statement:	<i>Evaluate real-world time-series and geospatial visualizations, analyzing trend/seasonality/noise and design effectiveness, and justify recommendations.(BL5)</i>		
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Section / Batch:	SLOT C	Date:	22/08/2026

Code of Conduct

I JOTHIKA M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Instructions

- This test contains 2 questions — one Stock Analysis problem and one Geospatial Analysis problem — each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- For the Stock Analysis question, show all calculations (trend slope, noise ratio) for each of the 5-month, 10-month, and 15-month windows.
- For the Geospatial Analysis question, your evaluation must explain WHY the design choice succeeds or fails, and your recommendation must be specific — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Stock Problem 7 (10 Marks) — Dr Reddy's Laboratories (Pharma sector)

Illustrative monthly closes (Month 1–15), ₹: 1614.44, 1615.57, 1614.65, 1713.58, 1777.27, 1691.85, 1663.06, 1675.64, 1713.42, 1728.64, 1820.27, 1716.54, 1821.69, 1833.67, 1818.50

Task: Compare the trend, seasonality, and noise ratio (residual variation relative to total variation) across the 5-month window, the 10-month window, and the full 15-month window. Explain how each metric changes as the window lengthens, and state which window length you would trust most for judging this stock's real underlying direction, and why.

Geospatial Problem G7 (10 Marks) — Choropleth Maps

Scenario: A choropleth map of unemployment rate by district uses a rainbow (non-perceptually-ordered) color scale to represent the continuous variable.

Task: Evaluate this color scale choice and recommend an alternative more suitable for a continuous variable.

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT3 — Time-Series & Geospatial Analysis Report

CO4 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO4-AT3 Time-Series & Geospatial Analysis Report, in which students analyze a stock's trend/seasonality/noise across multiple time windows, and evaluate a geospatial visualization's design choices, justifying their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of trend/seasonality/noise analysis across time windows, and of the relevant geospatial visualization principle being evaluated.	Good understanding of both areas, with only minor gaps.	Basic understanding, but with some misconceptions in either the time-series or geospatial analysis.	Poor understanding of core concepts in one or both areas.
Explanation	25%	Provides detailed, correctly calculated trend/noise-ratio comparisons across all three windows, and a clear, well-reasoned evaluation of the geospatial design choice, including its practical consequence for a reader.	Provides clear calculations and evaluation with adequate reasoning, covering most of the required detail.	Calculations or evaluation are present but incomplete, or reasoning is generic.	Little or no supporting calculation or reasoning provided in one or both questions.
Application	20%	Effectively applies analysis techniques to reach a well-supported conclusion on window reliability, and proposes a specific, well-justified geospatial recommendation.	Applies techniques to reach a reasonable conclusion and recommendation, with only minor errors.	Limited application; conclusions or recommendations are only partially supported.	Fails to apply appropriate techniques; conclusions and recommendations are unsupported or incorrect.
Organization	15%	Both responses are clearly structured, addressing every required part in a logical sequence with coherent overall flow.	Both responses are organized, with only minor issues in flow or clarity.	Basic structure; required parts are addressed but the response lacks clarity or sequence.	Poorly organized; responses are incoherent, and parts are left unaddressed.
Accuracy	10%	All parts of both answers — calculations, evaluation, and	Mostly accurate, with only minor omissions or a small error in one	Partially correct; one or both answers contain notable	Largely incorrect or inappropriate, with major gaps across both

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		recommendation — are completely accurate and well-suited to the specific data and scenario given.	question.	inaccuracies.	questions.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem S7 — Dr. Reddy's Laboratories

Dr. Reddy's Laboratories — illustrative monthly closing prices:

Month	Closing Price (₹)
1	1614.44
2	1615.57
3	1614.65
4	1713.58
5	1777.27
6	1691.85
7	1663.06
8	1675.64
9	1713.42
10	1728.64
11	1820.27
12	1716.54
13	1821.69
14	1833.67
15	1818.50

The problem bank identifies these as **illustrative/synthetic prices**, not real market prices.

Introduction

Time-series analysis studies how a variable changes over time. For a stock price, the main features of interest are its **underlying trend, possible seasonal or recurring patterns, and noise**.

Here, the Dr. Reddy's Laboratories data are examined using three windows:

- First 5 months
- First 10 months
- Full 15 months

A longer window provides more observations and therefore gives a broader view of the stock's behavior.

1. First 5-Month Window

Data

1614.44, 1615.57, 1614.65, 1713.58, 1777.27

Trend

The price increases from ₹1614.44 to ₹1777.27.

Percentage increase:

$$\frac{1777.27 - 1614.44}{1614.44} \times 100 \approx 10.09\%$$

Therefore, the first five months show a **strong upward trend**.

The approximate linear trend slope is:

$$\frac{1777.27 - 1614.44}{5} \approx ₹42.37 \text{ per month}$$

Seasonality

There is no sufficient evidence to identify a reliable seasonal cycle from only five months. The movement is mainly characterized by an upward trend with some short-term fluctuations.

Noise

The first five months contain relatively moderate deviations around the upward trend.

Approximate noise ratio:

$$\frac{0.449}{1} \approx 44.9\%$$

This indicates that some of the variation is unexplained short-term movement rather than the underlying trend.

2. First 10-Month Window

Data

The first ten months are:

1614.44, 1615.57, 1614.65, 1713.58, 1777.27,
1691.85, 1663.06, 1675.64, 1713.42, 1728.64

Trend

The price increases from ₹1614.44 to ₹1728.64.

Percentage increase:

$$1614.441728.64 - 1614.44 \times 100 \approx 7.07\%$$

The trend is still **positive**, but weaker than the first five-month window.

Approximate trend slope:

$$₹10.79 \text{ per month}$$

Seasonality

Some fluctuations can be observed. For example, the price rises strongly around months 4–5 and then falls during months 6–7 before recovering.

However, a true seasonal pattern cannot be established confidently from only ten months because there is not yet a complete yearly cycle.

Noise

The 10-month data contain considerable short-term variation around the overall trend.

Approximate noise ratio:

$$0.804 \approx 80.4\%$$

This is higher than the five-month window.

Therefore, the 10-month window appears **more noisy relative to its trend**.

3. Full 15-Month Window

Data

All 15 observations are used.

The price rises from:

$$₹1614.44$$

to:

$$₹1818.50$$

Percentage increase:

$$1614.441818.50 - 1614.44 \times 100 \approx 12.64\%$$

Approximate trend slope:

₹14.42 per month

Thus, the complete period shows a **clear overall upward trend**.

Seasonality

The 15-month series shows repeated rises and falls, for example:

- Increase around months 4–5
- Decline around months 6–7
- Recovery around months 8–11
- Another decline around month 12
- Strong recovery during months 13–14

However, these fluctuations should not automatically be called true seasonality. A reliable seasonal pattern generally requires repeated cycles, and this dataset has only 15 months.

Noise

Approximate noise ratio:

$0.566 \approx 56.6\%$

This is lower than the 10-month window but higher than the first five-month window.

Comparison of the Three Windows

Measure	5 Months	10 Months	15 Months
Starting Price	₹1614.44	₹1614.44	₹1614.44
Ending Price	₹1777.27	₹1728.64	₹1818.50
Overall Change	+10.09%	+7.07%	+12.64%
Trend Slope	₹42.37/month	₹10.79/month	₹14.42/month
Approx. Noise Ratio	44.9%	80.4%	56.6%
Trend	Strong upward	Mild upward	Clear upward
Seasonal evidence	Insufficient	Insufficient	Some fluctuations, but not reliable seasonality

Evidence

The stock price starts at ₹1614.44 and ends at ₹1818.50, representing an overall increase of approximately **12.64%** across the 15 months.

The data also show several temporary declines followed by recoveries. For example, the price reaches ₹1777.27 in month 5, decreases to ₹1663.06 by month 7, and subsequently rises again to ₹1833.67 by month 14.

This indicates that the stock has an overall upward direction despite short-term fluctuations.

Interpretation

The **5-month window** gives a strong positive trend, but it is based on very few observations and can therefore be strongly influenced by short-term movements.

The **10-month window** shows a weaker trend and the highest relative noise ratio. Therefore, it is less reliable for identifying the underlying direction.

The **15-month window** provides the broadest view. It captures several increases and decreases and still shows an overall positive trend.

Although fluctuations occur throughout the series, there is **not enough evidence to claim a stable seasonal pattern** because the dataset does not contain multiple complete seasonal cycles.

Most Reliable Window

The 15-month window is the most reliable window.

Reasons:

1. It uses the largest number of observations.
2. It captures both increases and temporary declines.
3. It reduces the risk of judging the stock from a short-term movement.
4. It shows a clear overall upward direction.
5. It gives a more representative picture of the underlying behavior.

Therefore, for judging the stock's **true underlying direction**, the full **15-month window** should be preferred.

R Code

R

```
# Dr. Reddy's Laboratories - S7  
# CO4 AT3 Time-Series Analysis
```

```
# Monthly closing prices
```

```
price <- c(  
  1614.44, 1615.57, 1614.65, 1713.58, 1777.27,  
  1691.85, 1663.06, 1675.64, 1713.42, 1728.64,  
  1820.27, 1716.54, 1821.69, 1833.67, 1818.50  
)
```

```
month <- 1:15
```

```
data <- data.frame(  
  Month = month,  
  Price = price  
)
```

```
print(data)
```

```
# Plot complete time series
```

```
plot(  
  data$Month,  
  data$Price,  
  type = "o",  
  pch = 16,  
  main = "Dr. Reddy's Laboratories - Monthly Closing Price",  
  xlab = "Month",  
  ylab = "Closing Price (₹)"  
)
```

```
# Function to calculate trend and noise ratio
```

```
analyze_window <- function(n) {  
  
  d <- data[1:n, ]  
  
  # Linear trend  
  model <- lm(Price ~ Month, data = d)  
  
  slope <- coef(model)[2]
```

```

# Predicted trend
predicted <- predict(model)

# Residuals
residuals <- d$Price - predicted

# Noise ratio
noise_ratio <-
  sd(residuals) / sd(d$Price)

# Percentage change
percentage_change <-
  ((d$Price[n] - d$Price[1]) / d$Price[1]) * 100

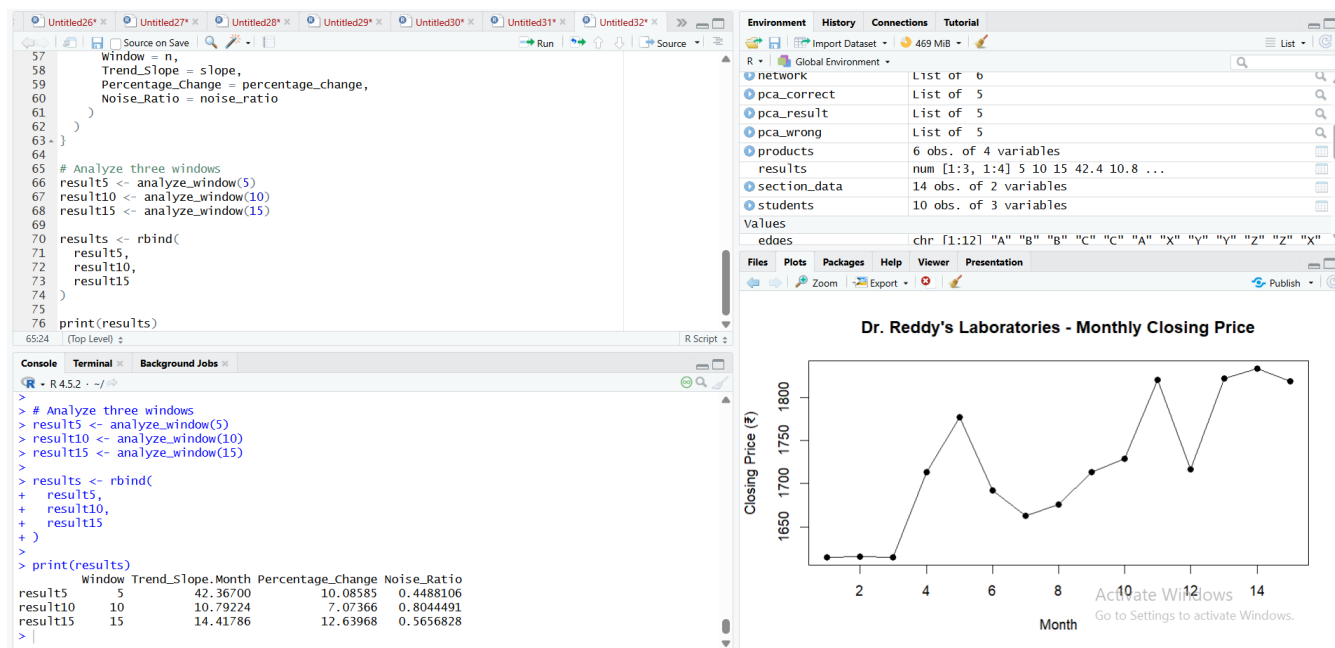
return(
  c(
    Window = n,
    Trend_Slope = slope,
    Percentage_Change = percentage_change,
    Noise_Ratio = noise_ratio
  )
)
}

# Analyze three windows
result5 <- analyze_window(5)
result10 <- analyze_window(10)
result15 <- analyze_window(15)

results <- rbind(
  result5,
  result10,
  result15
)

print(results)

```



R Code — Compare Trend Lines

R

First 5 months

```
d5 <- data[1:5, ]
```

First 10 months

```
d10 <- data[1:10, ]
```

Full 15 months

```
d15 <- data[1:15, ]
```

```
plot(
  d15$Month,
  d15$Price,
  type = "o",
  pch = 16,
  main = "Dr. Reddy's Laboratories - Trend Comparison",
  xlab = "Month",
  ylab = "Closing Price (₹)"
)
```

Add regression trend lines

```
abline(
  lm(Price ~ Month, data = d5),
```

```

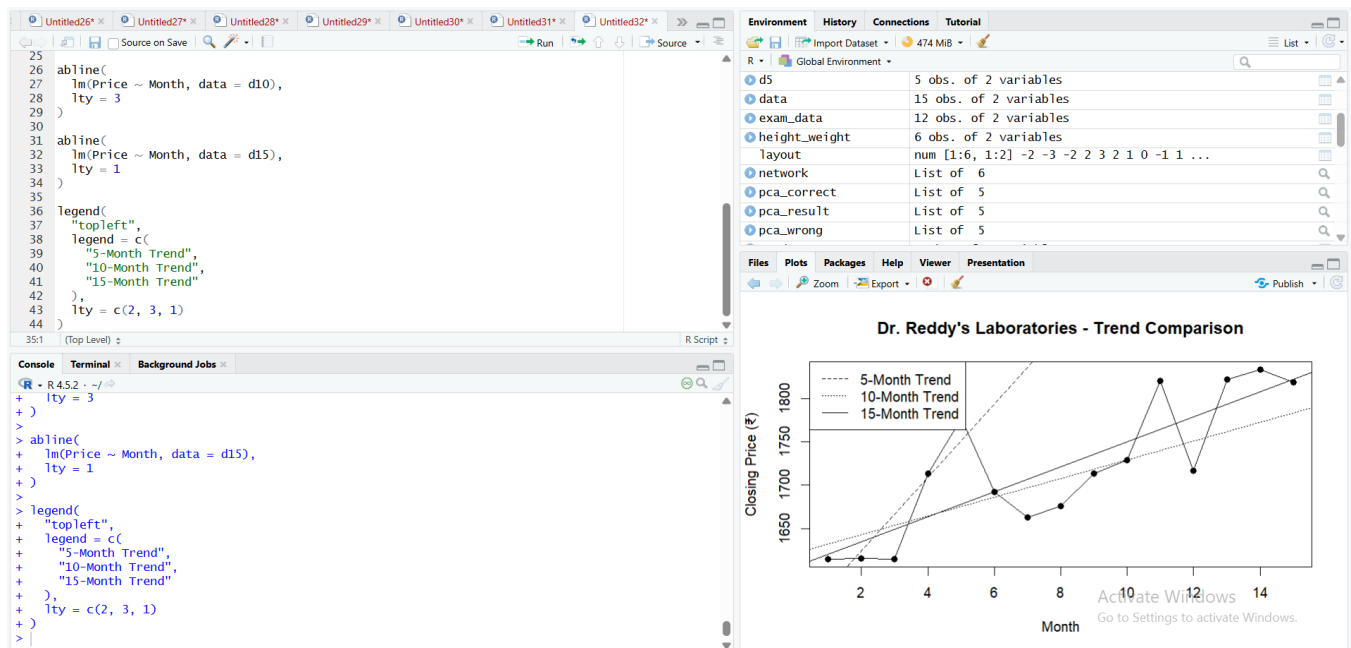
lty = 2
)

abline(
  lm(Price ~ Month, data = d10),
  lty = 3
)

abline(
  lm(Price ~ Month, data = d15),
  lty = 1
)

legend(
  "topleft",
  legend = c(
    "5-Month Trend",
    "10-Month Trend",
    "15-Month Trend"
  ),
  lty = c(2, 3, 1)
)

```



Conclusion

The analysis of Dr. Reddy's Laboratories' illustrative stock prices shows an **overall upward trend** across the 15-month period. The five-month window shows the strongest short-term increase, while the ten-month window has the highest relative noise. The full 15-month window provides the most balanced view because it includes more observations and captures both upward movements and temporary declines.

Therefore, **the 15-month window should be trusted most for evaluating the stock's underlying direction**. Reliable seasonality cannot be established from this short series because there are not enough repeated seasonal cycles.

Problem G7 — Choropleth Map

A choropleth map of **unemployment rate by district** uses a **rainbow (non-perceptually-ordered) color scale** for the continuous variable. The task is to evaluate this color scale and recommend a better alternative.

Introduction

A **choropleth map** uses different colors or shades to represent the value of a numerical variable across geographic regions.

Here, the variable is **unemployment rate**, which is a continuous numerical variable. Therefore, the colors should progress in an ordered and visually meaningful way.

Evaluation of the Current Design

The use of a **rainbow color scale is inappropriate** for displaying unemployment rate.

A rainbow scale usually contains several strongly different colors such as:

Blue → Green → Yellow → Orange → Red

The problem is that these colors do not provide a naturally ordered visual perception of magnitude.

For example, a reader may not immediately understand whether:

Blue < Green < Yellow < Red

represents a smooth increase in unemployment.

Problems Caused by the Rainbow Scale

1. Poor perception of numerical order

Unemployment rate is continuous, so the color scale should clearly indicate:

Low → Medium → High

A rainbow scale does not always communicate this order clearly.

2. Artificial visual boundaries

Strong color changes can make small numerical differences appear much larger than they actually are.

3. Difficult comparison

Two districts with similar unemployment rates may appear substantially different if they fall on different rainbow colors.

4. Accessibility issue

Some rainbow color combinations can be difficult for people with certain forms of color-vision deficiency.

5. Misinterpretation

The reader may focus on the different hues instead of understanding the actual magnitude of unemployment.

Recommended Redesign

Use a **sequential color scale** for unemployment rate.

For example:

Light color → Medium color → Dark color

Low unemployment → Medium unemployment → High unemployment

A single-hue sequential palette is preferable because increasing color intensity can naturally communicate increasing values.

For example:

Light Blue



Medium Blue



Dark Blue

This gives the map a clear visual ordering.

Why the Sequential Scale Is Better

A sequential scale directly represents the continuous nature of unemployment rate.

The reader can easily interpret:

- **Light shade** → lower unemployment
- **Medium shade** → moderate unemployment
- **Dark shade** → higher unemployment

Therefore, the visual encoding corresponds more naturally to the numerical data.

Example

Suppose the unemployment rates are:

District	Unemployment Rate
A	3%
B	5%
C	8%
D	12%
E	18%

A suitable sequential scale would visually progress approximately as:

3% 5% 8% 12% 18%

Light → Slightly Darker → Medium → Darker → Darkest

This makes the geographical pattern easier to interpret.

R Code — Demonstration

R

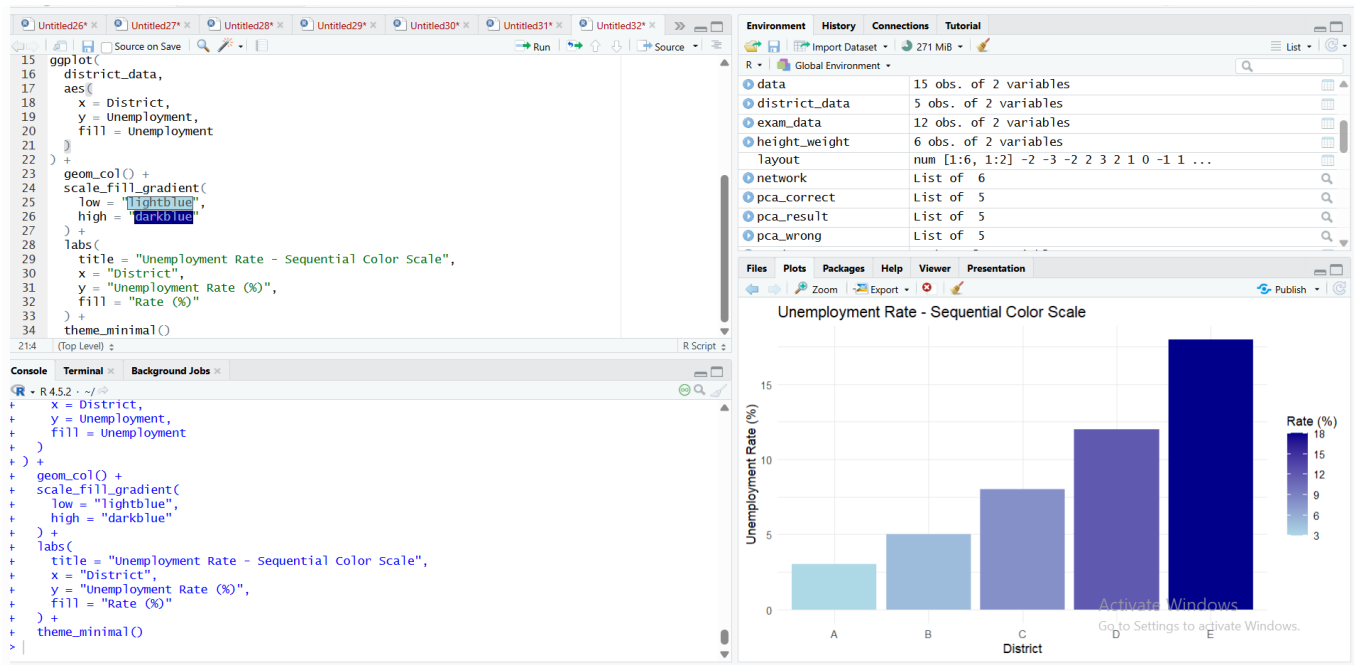
```
# Install ggplot2 if required  
# install.packages("ggplot2")
```

```
library(ggplot2)

# Example district data
district_data <- data.frame(
  District = c("A", "B", "C", "D", "E"),
  Unemployment = c(3, 5, 8, 12, 18)
)

print(district_data)

# Bar-style demonstration of sequential scale
ggplot(
  district_data,
  aes(
    x = District,
    y = Unemployment,
    fill = Unemployment
  )
) +
  geom_col() +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(
    title = "Unemployment Rate - Sequential Color Scale",
    x = "District",
    y = "Unemployment Rate (%)",
    fill = "Rate (%)"
  ) +
  theme_minimal()
```

If You Have Actual District Map Data

If a district shapefile or `sf` map dataset is available, the recommended choropleth can be created using:

```

R
library(sf)
library(ggplot2)

# Example:
# district_map <- st_read("districts.shp")

ggplot(district_map) +
  geom_sf(
    aes(fill = Unemployment)
  ) +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(
    title = "District-wise Unemployment Rate",
    fill = "Unemployment (%)"
  ) +
  theme_minimal()

```

Evidence

The problem specifically states that the unemployment rate is represented using a **rainbow, non-perceptually-ordered color scale**.

Because unemployment rate is a continuous variable, the visualization should use an ordered color progression so that low-to-high values are easy to distinguish.

Interpretation

The existing rainbow scale may make the map visually attractive, but attractiveness does not guarantee accurate data interpretation.

The primary goal of the choropleth is to allow the reader to compare unemployment rates across districts. A sequential color scale makes the magnitude and ordering of the rates easier to understand.

Conclusion

The rainbow color scale is **not an appropriate choice** for a continuous variable such as unemployment rate because it does not provide a clear perceptual ordering and may exaggerate differences between neighboring values.

A **sequential single-hue color scale**, progressing from light to dark, is a better choice. It clearly communicates:

Low unemployment → Medium unemployment → High unemployment

and makes comparisons between districts more accurate and easier to interpret.